# Algebra: Partial Fractions & Division

## Section 1: Polynomial Division & Improper Fractions

**1** Divide $2x^3 - 3x^2 + 4x - 5$ by $(x - 2)$ to find the quotient and remainder.

**a.** Quotient $= 2x^2 + x + 6$, Remainder $= 7$

**b.** Quotient $= 2x^2 - 7x + 18$, Remainder $= -41$

**c.** Quotient $= 2x^2 + x + 6$, Remainder $= -7$

**d.** Quotient $= 2x^2 + 7x + 18$, Remainder $= 31$

**2** Express the improper rational function $\frac{x^2 + 5x + 7}{x + 2}$ in the form $Ax + B + \frac{C}{x + 2}$.

**a.** $x + 3 + \frac{1}{x + 2}$

**b.** $x + 3 + \frac{7}{x + 2}$

**c.** $x + 7 - \frac{7}{x + 2}$

**d.** $x + 2 + \frac{3}{x + 2}$

**3** When $P(x) = x^4 + ax^2 + bx - 4$ is divided by $(x - 1)$, the remainder is 2; when divided by $(x + 2)$, the remainder is $-8$. Find $a$ and $b$.

**a.** $a = 1, b = 4$

**b.** $a = 2, b = 3$

**c.** $a = -1, b = 6$

**d.** $a = 3, b = 2$

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## Section 2: Partial Fractions with Linear & Repeated Factors

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**4** Express $\dfrac{7x-1}{(x-1)(x+2)}$ in partial fractions.

**a.** $\dfrac{2}{x-1}+\dfrac{5}{x+2}$

**b.** $\dfrac{5}{x-1}+\dfrac{2}{x+2}$

**c.** $\dfrac{6}{x-1}+\dfrac{1}{x+2}$

**d.** $\dfrac{7}{x-1}-\dfrac{1}{x+2}$

**5** What is the correct partial fraction decomposition template for $\dfrac{3x^2+1}{(x-2)^2(x+3)}$?

**a.** $\dfrac{A}{x-2}+\dfrac{B}{(x-2)^2}+\dfrac{C}{x+3}$

**b.** $\dfrac{A}{(x-2)^2}+\dfrac{B}{x+3}$

**c.** $\dfrac{Ax+B}{(x-2)^2}+\dfrac{C}{x+3}$

**d.** $\dfrac{A}{x-2}+\dfrac{B}{x+3}$

**6** In the decomposition $\dfrac{4x}{(x-1)^2}=\dfrac{A}{x-1}+\dfrac{B}{(x-1)^2}$, find $A$ and $B$.

**a.** $A=4, B=4$

**b.** $A=4, B=0$

**c.** $A=0, B=4$

**d.** $A=2, B=4$

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## Section 3: Irreducible Quadratic Factors

**7** What is the correct partial fraction template for $\dfrac{2x+5}{(x-1)(x^2+4)}$?

**a.** $\dfrac{A}{x-1}+\dfrac{Bx+C}{x^2+4}$

**b.** $\dfrac{A}{x-1}+\dfrac{B}{x^2+4}$

**c.** $\dfrac{Ax+B}{x-1}+\dfrac{C}{x^2+4}$

**d.** $\dfrac{A}{x-1}+\dfrac{B}{x+2}+\dfrac{C}{x-2}$

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**8** In $\dfrac{7x^2+5x+13}{(x-1)(x^2+4)} = \dfrac{A}{x-1} + \dfrac{Bx+C}{x^2+4}$, find the value of $A$.

**a.** 5

**b.** 7

**c.** 2

**d.** 3

**9** For the decomposition in the previous question ($A = 5$), find $B$ and $C$.

**a.** $B = 2, C = 7$

**b.** $B = 2, C = -7$

**c.** $B = 7, C = 2$

**d.** $B = -2, C = 7$

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## Section 4: Binomial Series with Partial Fractions

**10** To expand $f(x) = \dfrac{7x-1}{(x-1)(x+2)}$ in ascending powers of $x$ for small $|x|$, which partial fraction form should be expanded?

**a.** $-\dfrac{2}{1-x} + \dfrac{5/2}{1+x/2}$

**b.** $\dfrac{2}{x-1} + \dfrac{5}{x+2}$

**c.** $\dfrac{2}{1-x} + \dfrac{5}{1+x/2}$

**d.** $-\dfrac{2}{1-x} + \dfrac{5}{1+x}$

**11** Find the constant term in the expansion of $\dfrac{7x-1}{(x-1)(x+2)}$ in ascending powers of $x$.

**a.** $\dfrac{1}{2}$

**b.** $-\dfrac{1}{2}$

**c.** $-2$

**d.** 5

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**12** State the radius of convergence for the expansion of $\frac{7x-1}{(x-1)(x+2)}$ in ascending powers of $x$.

**a.** $|x| < 1$

**b.** $|x| < 2$

**c.** $|x| < 1/2$

**d.** $|x| < 7$

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